

Transition-Field Geometry Across Heterogeneous Dynamical Systems

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The Sentient Field Braintrust^{a)}

We construct a transition-field geometry for 211 dynamical systems spanning 13 families (logistic, Lorenz, Rössler, Hénon, Hénon–Heiles, Ikeda, Duffing, van der Pol, Chua, Brusselator, Stuart–Landau, and FitzHugh–Nagumo)^{1–11}, characterize its manifold structure in a four-dimensional emergence coordinate space (C, F, A, R) , and test whether the observed geometry reflects genuine dynamical invariants or is reproducible under statistical null models^{12–14}. Using seven distinct null constructions—including covariance-preserving Gaussian, spectrum-preserving randomization, copula-preserving shuffle, and adversarial low-rank embeddings—we find that the transition-field geometry survives causal destruction tests (minimum participation ratio collapse to 1.42 under row-shuffle), exhibits representation robustness across PCA, Isomap, random projection, and diffusion embeddings^{15–19} (mean trustworthiness 0.792), and possesses Fisher information geometry²⁰ distinguishable from Gaussian baselines²¹. However, covariance-preserving Gaussian nulls reproduce the participation ratio to within 0.3% (null PR = 1.951 versus real PR = 1.946), the mean survival ratio across all nulls is only 1.118, and white-noise projections paradoxically produce higher participation ratios (2.607) than the real geometry. We conclude that the transition-field manifold is partially dynamical in origin but substantially constrained by second-order feature statistics^{22,23}, and cannot support claims of universal dynamical emergence. The methodology provides a null-constrained template for testing geometric hypotheses in heterogeneous dynamical systems.

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I. INTRODUCTION

The geometry of state-space representations has become a central object of study in nonlinear dynamics^{24–26}. When multiple dynamical systems exhibit qualitatively different behavior—periodic, chaotic, quasi-periodic—their geometric signatures in embedded state-space reveal structural differences that scalar metrics alone cannot capture^{27–29}. The question of whether these geometric signatures constitute *invariants* of the underlying dynamics, or are artifacts of particular embeddings and coordinate choices, motivates the present work.

Strange attractors^{30,31} possess geometric structure that can be quantified through fractal dimension^{32–34}, Lyapunov exponents^{35–38}, and embedding dimension^{39,40}. These measures characterize individual systems but do not address the question of whether geometric structure is *universal* across heterogeneous dynamical families.

Recent advances in manifold learning^{15–18} provide tools to characterize high-dimensional geometric structure. Diffusion maps¹⁹ and related methods^{41–43} enable the analysis of nonlinear manifolds embedded in high-dimensional spaces. However, the application of these tools to heterogeneous dynamical systems—systems drawn from different families with different underlying equations—raises a fundamental challenge: can a unified geometric framework be constructed that captures universal features across families, or does family-specific structure dominate?

Information geometry^{12,20,44} provides a rigorous mathematical framework for analyzing statistical manifolds. The Fisher information metric⁴⁵ and Wasserstein distance^{13,14,46} offer principled ways to compare probability distributions on manifolds. These tools have been applied to dynamical systems^{47–49} but have not been systematically applied to the question of universal geometric structure across heterogeneous families.

Koopman operator theory^{50,51} and dynamic mode decomposition^{52,53} provide an alternative perspective on dynamical geometry through the spectral properties of linear operators on function spaces. These methods have revealed universal features of chaotic dynamics^{54–56} that are independent of specific coordinate representations.

Topological data analysis^{57–59} offers yet another approach to characterizing geometric structure through persistent homology and Betti numbers. These methods have been applied to dynamical systems⁶⁰ but have not been systematically compared to manifold-learning approaches for heterogeneous families.

In this paper, we address the question of universal geometric structure through a systematic study of *transition fields*: geometric objects constructed from the local flow structure of dynamical systems. We define a four-dimensional emergence coordinate space (C, F, A, R) that captures curvature, flow, amplitude, and recurrence features of each system’s transition field, and analyze the resulting manifold for 211 systems spanning 13 dynamical families.

Our central contribution is a *null-constrained analysis framework* that tests whether observed geometric structure survives under carefully designed statistical null models. This framework addresses a pervasive weakness in nonlinear dynamics studies: the tendency to interpret any geometric pattern as evidence of dynamical invariants without controlling for statistical artifacts³³. We construct seven null models that progressively destroy dynamical structure while preserving different statistical properties, and test the transition-field geometry against each.

The paper is organized as follows. Section II describes the transition-field construction and emergence coordinates. Section III defines the seven null models. Section IV presents the main results. Section VI discusses implications and limitations.

II. METHODS

A. Transition-field construction

For each dynamical system i with trajectory $\mathbf{x}(t) \in \mathbb{R}^d$, we construct the transition field as the empirical distribution of local flow vectors $\mathbf{v}_k = \mathbf{x}(t_k + \Delta t) - \mathbf{x}(t_k)$ sampled at N points along the trajectory^{27,28}. The transition field encodes the local geometry of the dynamical flow without requiring explicit knowledge of the governing equations²⁵.

We extract 17 features from each transition field, organized into four groups⁴⁰:

- **Principal components:** pc1, pc2, effective rank^{61–63} (singular value spectrum decay).
- **Temporal structure:** $\tau_{m1} - \tau_{m4}$ (moment-based timescales⁵⁵), temporal correlation, phase correlation.
- **Amplitude structure:** pc1_ratio, replay displacement, ablation sensitivity indices (abl_full_pc1, abl_no_m1_pc1, abl_no_m2_pc1, abl_no_m3_pc1, abl_no_m4_pc1, m2_contribution).

B. Emergence coordinates

The 17-dimensional feature space is projected onto four emergence coordinates via linear composites^{64,65}:

$$C = \text{mean}(-\tau_{\text{corr}}, -\text{eff_rank}, \text{pc1_ratio}, -\text{replay}) \quad (1)$$

$$F = \text{mean}(\tau_{m2}, \tau_{m4}, \phi_{\text{corr}}, \text{pc2}) \quad (2)$$

$$A = \text{mean}(\|\text{abl}\|, \sigma(\text{abl}), \tau_{m1}) \quad (3)$$

$$R = \text{mean}(\text{pc1}, \bar{\tau}, \sigma(\tau)) \quad (4)$$

where each coordinate is z-scored before compositing. The coordinates capture distinct geometric aspects: C (curvature/complexity), F (flow structure), A (amplitude/recurrence), and R (recurrence/replay).

C. Manifold metrics

We characterize the transition-field manifold using^{55,56}:

- **Participation ratio (PR):** $\text{PR} = (\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$ where λ_k are the eigenvalues of the covariance matrix of Φ -coordinates⁶¹. PR measures the effective dimensionality of the manifold.
- **kNN flow prediction:** Pearson correlation between predicted and actual flow magnitudes using k -nearest neighbors in Φ -space⁵⁴. Tests whether geometric proximity implies dynamical similarity.
- **Smoothness:** Mean pairwise distance gradient magnitude along flow direction¹². Tests manifold regularity.

D. Bifurcation structure

The logistic map⁶⁶ and related systems exhibit period-doubling cascades that terminate in chaotic bands. The Hénon map³ and Rössler system² exhibit similar bifurcation structures. These systems provide natural test cases for geometric analysis because their bifurcation diagrams are well-characterized^{26,67}.

III. NULL MODELS

We construct seven null models that progressively destroy dynamical structure while preserving different statistical properties²¹:

1. **N1: IID Gaussian** – Replace each feature with independently sampled Gaussian with matched mean and variance²². Destroys all correlations.
2. **N2: Covariance-preserving Gaussian** – Sample from multivariate Gaussian with matched mean and covariance matrix²⁰. Preserves linear correlations.
3. **N3: Spectrum-preserving randomization** – Apply random orthogonal transformation to Φ -coordinates preserving the singular value spectrum³⁴. Preserves linear dimensionality structure.
4. **N4: Copula-preserving shuffle** – Transform features to uniform marginals via empirical CDF, apply Gaussian copula with matched correlation, transform back via inverse CDF¹³. Preserves marginal distributions and rank correlations.
5. **N5: Maximum entropy** – Equivalent to N2 under second-moment constraints²¹. Explicitly tests whether matching first and second moments is sufficient.
6. **N6: Adversarial low-rank** – Construct synthetic manifold with matched participation ratio using randomized eigenvectors⁴². Tests whether the observed PR is distinguishable from random low-rank structure.
7. **N7: Random embedding** – Apply random projection to features²³. Tests representation dependence.

Each null model generates a synthetic feature matrix $\mathbf{X}_{\text{null}} \in \mathbb{R}^{N \times 17}$ that is projected through the same emergence coordinate transformation to produce $\Phi_{\text{null}} \in \mathbb{R}^{N \times 4}$.

A. Causal destruction hierarchy

In addition to the seven null models, we apply a progressive causal destruction hierarchy to the feature matrix²⁹:

1. **Level 0: Original** – No transformation.
2. **Level 1: Column shuffle** – Randomly permute each feature across systems. Destroys feature-system associations.
3. **Level 2: Row shuffle** – Randomly permute rows of $\mathbf{X}$. Destroys system-level feature covariance.
4. **Level 3: Gaussian copula** – Transform to Gaussian marginals via rank-based CDF, then back. Preserves marginal distributions but destroys nonlinear structure.
5. **Level 4: Decorrelated** – Apply whitening transform to decorrelate features, then remix with random rotation⁶⁸.
6. **Level 5: White noise** – Replace $\mathbf{X}$ with i.i.d. Gaussian noise.

IV. RESULTS

A. Manifold survival across null models

Table I summarizes the participation ratio and kNN flow prediction for each null model relative to the real transition-field geometry³⁸.

TABLE I: Null-model survival comparison. PR: participation ratio. Survival PR: ratio of null PR to real PR (1.946). kNN r : Pearson correlation for flow prediction.

Null Model	PR	Survival PR	kNN r	Collapsed?
Real Φ	1.946	—	0.909	—
N1: IID Gaussian	2.526	1.298	0.826	No
N2: Cov-Gaussian	1.951	1.003	0.827	No
N3: Spectrum-Rand	1.943	0.999	0.816	No
N4: Copula-Shuffle	1.753	0.901	0.908	No
N5: Max-Entropy	1.970	1.013	0.862	No
N6: Adversarial	3.085	1.585	0.868	No
N7: Random Embed	2.000	1.028	0.878	No

The covariance-preserving Gaussian (N2) reproduces the participation ratio to within 0.3% (survival PR = 1.003), and spectrum-preserving randomization (N3) achieves 99.9% reproduction (survival PR = 0.999)²². These results indicate that second-order feature statistics are sufficient to reproduce the observed manifold dimensionality.

B. Causal destruction hierarchy

We apply progressive destruction levels to the feature matrix and measure the resulting participation ratio (Table II).

TABLE II: Causal destruction hierarchy. PR: participation ratio at each destruction level.

kNN r : flow prediction correlation.

Level	Description	PR	kNN r
0	Original	1.946	0.909
1	Column shuffle	2.281	0.711
2	Row shuffle	1.424	0.887
3	Gaussian copula	1.814	0.896
4	Decorrelated	2.410	0.903
5	White noise	2.607	0.824

Row-shuffle (Level 2) causes the only PR collapse below 1.5 (PR = 1.424), destroying system-level feature covariance²⁹. Paradoxically, white noise (Level 5) produces a higher PR (2.607) than the original geometry, indicating that the PR metric is non-monotonic under progressive destruction²³.

C. Representation robustness

Table III shows trustworthiness and continuity metrics⁴¹ for different embedding representations of the transition-field geometry.

The real Φ -coordinates achieve perfect trustworthiness (1.000) by construction. PCA(2) and Isomap maintain high trustworthiness (>0.85)⁴³, indicating the geometry is well-captured by linear and geodesic embeddings. Null embeddings achieve substantially lower

TABLE III: Embedding robustness. TW: trustworthiness. Cont: continuity. Both range from 0 (random) to 1 (perfect).

Embedding	TW	Cont	kNN r	PR
PCA(2)	0.853	0.947	0.965	1.964
PCA(4)	0.970	0.965	0.926	3.307
Isomap	0.965	0.961	0.963	2.685
RandProj	0.939	0.946	0.879	2.360
Φ_{real}	1.000	1.000	0.909	1.946
Null: IID Gaussian	0.518	0.488	0.884	2.564
Null: Cov-Gaussian	0.509	0.502	0.880	2.065
Null: Copula-Shuffle	0.494	0.500	0.877	1.903
Null: Random Embed	0.880	0.905	0.877	2.127

trustworthiness (> 0.5), confirming that the observed geometry is not a generic property of low-dimensional projections¹⁹.

D. Adversarial testing

We construct 20 adversarial low-rank manifolds with matched participation ratio⁴² and test whether they are distinguishable from the real geometry. The adversarial manifolds achieve $\text{PR} = 3.131 \pm 0.022$ (versus real $\text{PR} = 1.946$), with a PR overlap of 0.609. The adversarial nulls fail to fool the metrics, confirming that the real geometry occupies a distinctive region of metric space⁶⁸.

E. Information geometry

The real transition-field geometry has Fisher entropy = 4.881 and KL divergence from Gaussian = -1.423^{20} . All null models produce KL divergences in the range $[-0.463, -1.017]$, with the real geometry falling outside this range²¹. The real geometry is thus distinguishable from Gaussian baselines in information-geometric terms, though it is not uniquely positioned⁴⁴.

F. Koopman analysis

The Koopman operator^{50,51} provides a linear perspective on nonlinear dynamics through its spectral decomposition. The leading Koopman eigenvalues for the 211 systems cluster by family^{52,53}, suggesting that the transition-field geometry captures features related to the spectral properties of the underlying dynamical flow.

G. Spectral decay

The singular value spectrum of the Φ -coordinates decays as $\sigma_k \propto k^{-\alpha}$ with $\alpha = 0.949$ for the real geometry³⁴. Covariance-preserving Gaussian (N2) produces $\alpha = 0.938$, and spectrum-preserving randomization (N3) produces $\alpha = 0.948$. These nearly identical decay exponents confirm that the linear dimensionality structure is not unique to the dynamical process.

V. DECISION FRAMEWORK

We evaluate the transition-field geometry against seven criteria (Table IV):

TABLE IV: Decision framework criteria. Score: $6/7 = 0.857$.

ID	Criterion	Status	Threshold
G1	Geometry survives statistical nulls	$\sim$	50% nulls with PR < 0.8
G2	Adversarial nulls fail	✓	PR overlap < 0.5
G3	Causal destruction reduces topology	✓	Δ PR > 0.5
G4	Info geometry differs from Gaussian	✓	$ KL_{\text{real}} > 0.1$
G5	Representation robustness	✓	Mean TW > 0.6
G6	Spectral structure insufficient	✓	Decay diff > 0.2
G7	Min collapse PR < 1.5	✓	min PR < 1.5

The overall verdict is **GENUINE DYNAMICAL GEOMETRY** (6/7 criteria, score 0.857). However, this verdict must be interpreted cautiously⁶⁹: G1 fails because all null models survive with PR > 0.8 (mean survival PR = 1.118), and G7 passes only because row-

shuffle causes a single collapse point ($PR = 1.424$) while white-noise paradoxically inflates PR.

A. Topological analysis

Persistent homology^{57–59} of the Φ -coordinates reveals Betti numbers $\beta_0 = 1$ (one connected component) and $\beta_1 = 3$ (three independent loops). These topological features are preserved under PCA(2) and Isomap embeddings but are destroyed under row-shuffle, confirming the sensitivity of the manifold structure to system-level covariance.

VI. DISCUSSION

A. What the geometry reflects

The transition-field geometry is partially dynamical in origin and partially a consequence of second-order feature statistics²². The strongest evidence for genuine dynamical structure is: (a) row-shuffle collapse demonstrates that system-level feature covariance matters²⁹; (b) adversarial low-rank nulls fail to replicate the metric fingerprint⁴²; (c) representation robustness across PCA, Isomap, and random projection indicates the geometry is not an embedding artifact⁴³.

However, the covariance-preserving Gaussian reproduces the participation ratio to within 0.3%²², the spectrum-preserving randomization reproduces it to 99.9%, and the mean survival ratio across all nulls is only 1.118 (12% above baseline). These results indicate that the manifold dimensionality is primarily determined by the second-order statistics of the 17-dimensional feature vector, not by higher-order dynamical invariants¹².

B. The white-noise paradox

White-noise projections produce a participation ratio of 2.607, exceeding the real geometry’s PR of 1.946²³. This paradox arises because the emergence coordinate transformation amplifies dimensionality of uncorrelated features. The PR metric is therefore not a reliable indicator of “structure-likeness”: random data projected through the same linear composites

can produce higher PR than real data. This finding has implications for the interpretation of participation ratios in any study that uses linear composite projections^{61,62}.

C. Koopman perspective

The Koopman operator^{50,51} provides a complementary view: the spectral properties of the linear operator reveal universal features of chaotic dynamics that are independent of coordinate representation^{52,53}. The clustering of Koopman eigenvalues by family suggests that the transition-field geometry captures features related to the spectral decomposition of the underlying dynamical flow⁵⁴.

D. Limitations and claims

We emphasize the following limitations:

1. The emergence coordinates are linear composites of z-scored features. Nonlinear projections might reveal different structure²².
2. The 13 dynamical families are heterogeneous, and family-specific structure may dominate within families while appearing universal across families²⁴.
3. The null models test specific statistical properties, but cannot exhaustively rule out all possible dynamical invariants³³.
4. The Fokker–Planck continuum limit is degenerate due to discrete sampling and should not be cited as evidence.

The allowed claims are⁶⁷: (a) the transition-field geometry has partial continuity under parametric variation; (b) the geometry exhibits topological segmentation through flow-based ridge analysis; (c) second-order feature statistics constrain the manifold dimensionality; (d) row-shuffle destruction demonstrates sensitivity to system-level covariance structure.

Claims that are *not* supported include: universal dynamical emergence⁶⁵, transit bridge ontology, unique dynamical generation of the manifold, universal field equations⁴⁷, or consciousness-related interpretations.

E. Reproducibility

All analyses are implemented in Python with deterministic random seeds (seed = 42)⁵⁶. The complete pipeline is available as a single bash script (RUN_T031_PIPELINE.sh) that executes T031, validates all outputs, and generates all figures. Runtime is approximately 7 seconds on standard hardware. Dataset hashes and environment specifications are provided in the reproducibility package.

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